

Real Estate System Requirements Checklist

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR1	User Registration & Login	High	User can create account, login/logout, reset password. Email unique.
FR2	Profile Management	High	User can update name, phone, profile photo, and plan. Changes saved instantly.
FR3	Property CRUD (Owner)	High	Owner can create, edit, publish/unpublish property with required fields.
FR4	Property Images	High	Owner can upload ≥ 1 image; set cover; images displayed in gallery.
FR5	Search & Browse	High	Guest/User can filter by type, location, price, status; see detail page.
FR6	Premium Visibility	High	Non-premium users cannot open Is_Premium_only=1 property; premium can.
FR7	Booking	High	User can request booking; Owner/Agent can confirm/decline; status transitions enforced.
FR8	Booking Conflicts	Medium	Two overlapping Confirmed bookings for same property → second rejected.
FR9	Payment	High	User can pay for booking; receipt generated; status shows Paid/Failed.
FR10	Refunds	Medium	Admin/Owner can trigger refund → refund record created.
FR11	Messaging	Medium	User can send/receive messages with property context; status = Sent/Read.

FR12	Reviews	High	User with Completed booking can leave 1-5 star rating + comment.
FR13	Loan Estimate	Medium	User can request estimate; monthly payment auto-calculated and saved.
FR14	Agent Console	Medium	Agent sees assigned properties, manages bookings, responds to messages.
FR15	CEO Console	Medium	CEO can add/disable Agents; view dashboards for bookings/payments.
FR16	Notifications	Medium	System sends email/SMS/push for bookings, payments, reviews, messages.
FR17	Audit Logs	High	Property/Booking/Payment status changes logged with actor + timestamp.
FR18	AR House View	Medium	User can open a property in AR mode via mobile; 3D model anchored and scaled.
FR19	VR House Tour	Medium	User can view property in VR mode; navigation via hotspots; supports WebXR/headsets.
FR20	3D Model Upload	Medium	Owner/Agent can upload .glb/.fbx model; system validates format/size.
FR21	3D Model Preview	Medium	Property detail page shows embedded 3D viewer; rotate, zoom, fullscreen.
FR22	AR/VR Premium Gate	Low	Premium users get early/extended access; free users see demo only.

Non-Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
NFR1	Performance	High	p95 page load $\leq 2.5s$ (public), $\leq 3.5s$ (auth); API search $\leq 300ms$.
NFR2	Scalability	High	Supports 100k users, 20k properties, 1M images without redesign.
NFR3	Availability	High	99.9% uptime (public); 99.5% back-office; DR restore ≤ 60 min.
NFR4	Security	High	HTTPS/TLS; password hashing; CSRF tokens; RBAC enforced.
NFR5	Payments Compliance	High	Card data tokenized; PCI SAQ-A validated.
NFR6	Privacy	High	User delete $\rightarrow$ PII removed within 30 days; reviews/messages retention $\leq 24m$.
NFR7	Accessibility	Medium	WCAG 2.1 AA; mobile-first; tap targets $\geq 44px$.
NFR8	Observability	Medium	Logs with correlation IDs; alerts on 5xx $>2\%$; DB CPU $>80\%$.
NFR9	Data Integrity	High	Price >0 ; at least 1 cover image before Active; bookings non-overlap.

NFR10	Maintainability	Medium	≥70% unit test coverage; CI/CD pipeline with lint + security scan.
NFR11	Portability	Low	Supports last 2 major browsers; ANSI SQL compliance.
NFR12	Performance (3D)	High	3D models load ≤5s broadband, ≤10s 4G; frame rate ≥30fps.
NFR13	Compatibility (AR/VR)	High	AR works on ARKit/ARCore; VR works with WebXR + headsets.
NFR14	Storage & CDN	Medium	3D assets ≤200MB; stored in CDN/object storage; cached globally.
NFR15	Security (AR/VR)	Medium	3D uploads scanned; only Owners/Agents can upload.
NFR16	Accessibility (Fallback)	Medium	If AR/VR unsupported, fallback to gallery or 360° photos.